

Aero 402: Bipropellant Thruster Experiment

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This abstract provides an overview of bipropellant thrusters, addressing their performance characteristics within the context of spacecraft propulsion systems. The primary objective of this research is to determine the performance characteristics of ethanol-oxygen bipropellant thrusters, investigated through the utilization of Cal Poly's AARLITE Bipropellant Thruster Experimental Test Setup. The experimental setup involved the use of a load cell and National Instrument DAQ for recording thrust measurements. Our findings reveal that there was a drop in combustion pressure during the burns in which the specific impulse resulted in a low value of 6.8228 seconds insinuating lack of efficiency in converting the propellant into thrust. Thus, providing important insights into how the performance of ethanol-oxygen bipropellant thruster can be effected by its test set up. In conclusion, this research significantly contributes to the field of spacecraft propulsion by offering a detailed examination of bipropellant performance. These findings have potential applications in a diverse range of missions, particularly those involving large satellites and spacecraft.

Nomenclature

A	=	Area of cross section
F	=	Thrust
I_{sp}	=	Specific Impulse
$\dot{m}_{fuel}$	=	Fuel Mass Flow rate
$\dot{m}_{oxidizer}$	=	Oxidizer Mass Flow rate
O/F	=	Oxidizer to Fuel Ratio
P	=	Pressure
r	=	Radius
V	=	Velocity
$\dot{V}$	=	Volume flow rate
ρ	=	Density

I. Introduction

BIPROPELLANT rocket engines operate through the controlled combination of initially separate liquid propellants and oxidizers within the combustion chamber, ignited to initiate propulsion. The injector plate, upon injection, is meticulously designed to facilitate the precise convergence of these two chemicals, resulting in atomization and the formation of reactants necessary for combustion. To ensure safety and efficiency, it is imperative to prevent premature mixing and reaction of these chemicals outside of the combustion chamber. This necessitates the implementation of a sophisticated feed system for both fuel and oxidizer, effectively mitigating back-flow concerns.

The majority of liquid propellant rockets rely on bipropellant systems, chosen based on mission-specific requirements, operating conditions (including vacuum scenarios), orbital parameters, and emission considerations. Notable examples of bipropellant rockets include SpaceX's Falcon family³ and engines manufactured by Blue Origin⁴. Given their widespread use in the aerospace industry, a comprehensive understanding of the characteristics and performance of bipropellant rocket systems is paramount.

The AARLITE Bipropellant Thruster exemplifies this category, employing a pressurized feed system to deliver liquid methanol fuel via a nitrogen gas tank, while relying on self-pressurized gaseous oxygen as the oxidizer. This laboratory endeavor seeks to leverage the AARLITE Bipropellant Thruster to enhance students' comprehension of the operational dynamics and performance attributes inherent to bipropellant rocket engines. In the subsequent sections, this report will delve into critical parameters including chamber pressure, exit pressure, nozzle expansion area, and thrust.

II. Test Setup

The AARLITE Bipropellant Thruster depicted in figure1 was employed in this experiment, utilizing gaseous oxygen as the oxidizer and liquid ethanol (CAS 64-17-5) as the fuel. The schematic illustrated in figure2 corresponds to the actual setup depicted in figure1. To pressurize the propellant, nitrogen from a stored cylinder was utilized, while the oxidizer, in its gaseous state, was stored in a separate cylinder. Water served as the coolant to monitor the heat rejection of the rocket. For measurements of different parameters, a load cell was employed to capture the force exerted by the thruster during the static fire. Additionally, the pressure gradient across the flow orifice was measured using pressure transducers. The analog voltage signals from the transducers were routed to a National Instruments 6036E data acquisition card, which converted them into digital data for reading and recording on the computer in the control room of the propulsion laboratory.



Fig. 1 Bipropellant Thruster

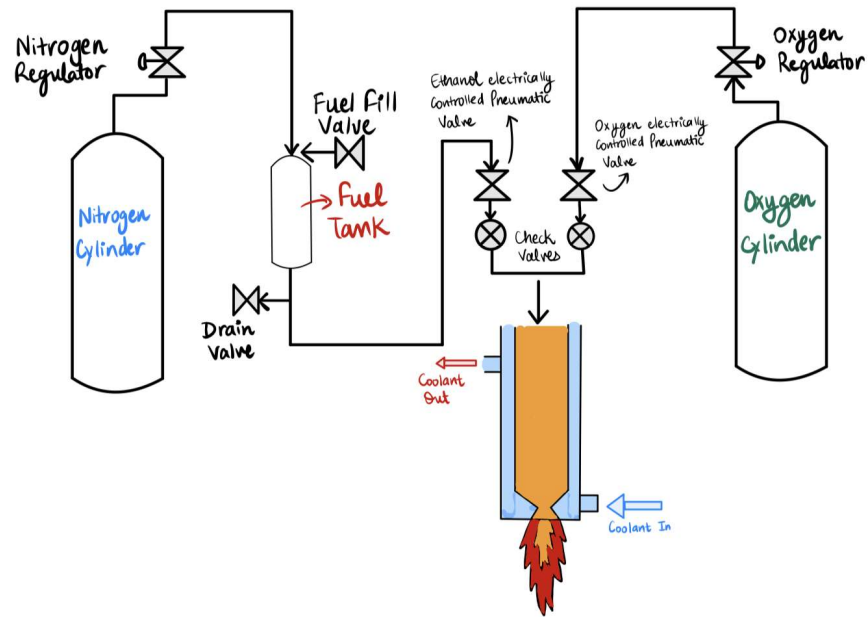


Fig. 2 AARLITE Bipropellant Thruster Schematic Test Setup

III. Methodology

A. Procedures

To ensure a safe and controlled test environment, several critical steps were taken before initiating the experiments. The team began by familiarizing themselves with the setup's components and assigning specific roles. Additionally, all

team members were equipped with essential protective gear, including ear mufflers and protective glasses. The test cell was meticulously cleared of any loose objects, and warning signs were prominently displayed outside the laboratory. Moreover, everyone was thoroughly briefed on the evacuation protocol in case of an emergency. Prior to conducting the cold run, a deviation from the lab manual's water flow procedure was necessitated by identified leaks. As an alternative, a makeshift coolant was ingeniously improvised using water bottles. The implications of this coolant system adjustment will be further addressed in the subsequent error section of this report. To prepare the fuel tank, a careful process was undertaken.

Initially, the nitrogen cylinder's pressure was finely adjusted to ensure minimal resistance during the unscrewing of the regulator, effectively achieving a pressure close to zero. A glass fuel beaker was strategically positioned beneath the fuel drain valve to eliminate any residual fuel in the tank. This procedure was facilitated using both the fuel drain and fuel fill valves. All readings on the lab computer were zeroed out, particularly those of the load cell and pressure transducers. This step was taken after confirming that the pressure values were within the acceptable range of ± 10 psi, and the load cell remained within ± 511 bf. To progress with the test, the pressures of the nitrogen and oxygen cylinders were set to 600 psi for the initial run, followed by 500 psi for the subsequent two runs. The air supply hose was then securely connected to the pneumatic valves. In order to nullify the nitrogen and oxygen pressures, their respective valves were enabled using the control box fig 3, ensuring that no sound was discernible.

With the fire hose unplugged, the ethanol was carefully introduced into the fuel tank. This process commenced with the depressurization of the fuel line via the fuel drain valve, followed by the controlled filling of the fuel. Crucially, the fuel handler maintained grounding throughout, and the nitrogen line leading into the fuel tank was positioned above to prevent any potential backflow. Upon securing the air supply for the pneumatic valves, preparations were made to initiate the thruster firing sequence. Through the control box, fuel and oxygen were enabled, and the actuators were subsequently triggered to record thruster data during firing.

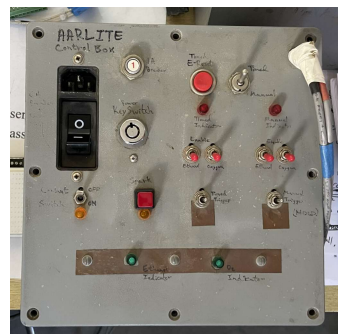


Fig. 3 Control Box

A total of three runs were successfully completed, although challenges were encountered in managing the makeshift coolant system. Further details regarding these difficulties will be discussed in the subsequent error section of this report. Following the conclusion of all runs, the prescribed rocket turnoff and closeout procedure outlined in the lab manual was adhered to, underscoring the team's commitment to ensuring the safety of both equipment and the laboratory environment.



(a) Nitrogen Tank w/ Pressure Gage



(b) Oxygen Tank w/ Presure Gage

B. Equations

Utilizing the data retrieved from the National instrument DAQ during the lab session. The data was uploaded to MATLAB (refer to the code in the appendix) to further refine the data and derive specific parameters stated in Table 1.

In order to derive the volumetric flow rate and subsequently obtain the mass flow rate of both the oxidizer and fuel, several steps are taken. It is assumed, for these calculations, that the flow is incompressible in the venturi. Starting with Equation 1, the pressure difference (ΔP) is determined by taking the difference between the low pressure values, measured from the throat, and the high pressure values, measured at the inlet of the venturi. The density of the oxidizer is 0.8457 slug/ft^3 , and for the ethanol fuel, it is 1.5295 slug/ft^3 . When inputting pressure values in lbf/ft^2 and density in slug/ft^3 , the resulting units for equation 1 are in ft/s .

$$V = \sqrt{\frac{2 \cdot \Delta P}{\rho}} \quad (1)$$

After obtaining the velocity of each of the oxidizer and fuel flows, we input the area of the inlet, measured in ft^2 . The diameter of the inlet, as given in the lab manual, is 0.0103 ft [1]. Using Equation 2, this results in an area of $3.3545 \times 10^{-4} \text{ ft}^2$ for both the oxygen and ethanol inlets, as stated in the lab manual for the thruster pressure feed design.

$$A = \pi \cdot r^2 \quad (2)$$

Using the area obtained in Equation 2, Equation 3 is used to find the volumetric flow rate of each propellant using the area and velocity of each respective fuel and oxidizer.

$$\dot{V} = A \cdot V \quad (3)$$

Once the volumetric flow rate is defined from Equation 3, Equation 4 utilizes the volumetric flow rate and the

densities to find the mass flow rate for the fuel and oxidizer.

$$\dot{m} = \rho \cdot \dot{V} \quad (4)$$

In addition to the mass flow rate being an important parameter of the system, the oxidizer-to-fuel ratio is crucial to determine if combustion is complete comparing it to the stoichiometric mixture ratio. Using both the oxidizer and fuel mass flow rates in Equation 5 yields the O/F ratio.

$$\frac{O}{F} = \frac{\dot{m}_{\text{oxidizer}}}{\dot{m}_{\text{fuel}}} \quad (5)$$

To further study the efficiency of the propellant converting into thrust, we derive the specific impulse using Equations 6 and 7. First, using Equation 6, the total mass flow rate is computed by adding the oxidizer and fuel values.

$$\dot{m}_{\text{total}} = \dot{m}_{\text{oxidizer}} + \dot{m}_{\text{fuel}} \quad (6)$$

Following this, Equation 7 is used to find the specific impulse by dividing the thrust, lbf, by 32.2 lbf to convert it to slug, then dividing the thrust, now slug, by the total mass flow rate, slug/s. This results in obtaining the correct units of seconds for I_{sp} .

$$I_{\text{sp}} = \frac{\frac{F}{32.2\text{lbf}}}{\dot{m}_{\text{total}}} \quad (7)$$

IV. Analysis & Results

A. Pressure & Thrust Data

To delve deeper into the traits of a bi-propellant thruster, we examine the data acquired from the runs. In the initial run, our focus is on pressure measurements. For further context, in all the pressure plots presented in this report for the fuel and oxidizer, the "low pressure" is measured at the throat, while the "high pressure" is measured at the inlet of the venturi. The pressure at the inlet of the venturi is typically higher than at the throat due to the principle of fluid dynamics. As the fluid flows through the converging section (throat) of the venturi, its velocity increases, following the continuity equation. According to Bernoulli's principle, an increase in fluid velocity corresponds to a decrease in pressure. Consequently, at the throat, where the fluid velocity is highest, the pressure is lowest.

Figure 5 illustrates a notable occurrence: the oxidizer pressure decreases during the burn. This is on par with Bernoulli's principle as stated previously, as the thruster during the burn increases the velocity of the system. Simultaneously, fuel pressures exhibit an initial drop but subsequently stabilize near their initial values. This data suggests potential concerns within the propulsion system, with fuel leakage being a likely issue.

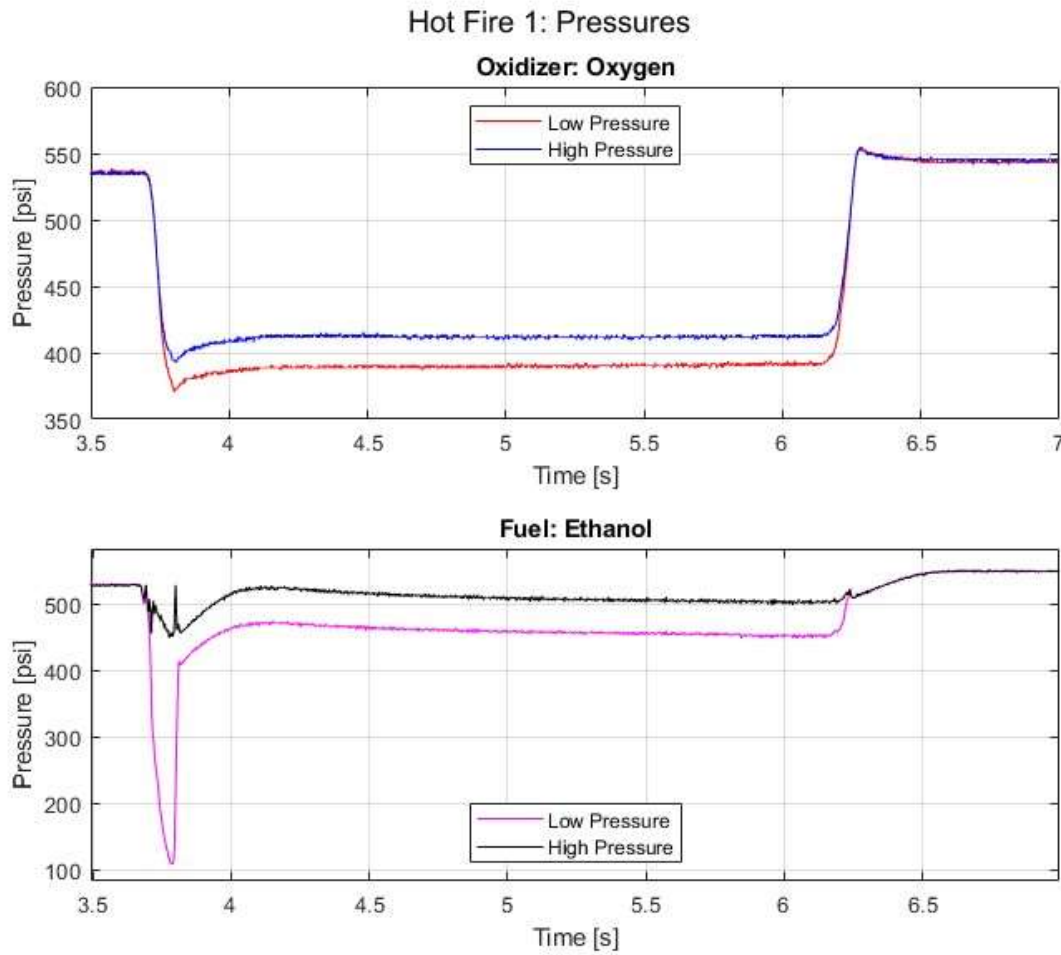


Fig. 5 First Run Pressure Data of Propellant

In the analysis of thrust and chamber pressure data, Figure 6, a critical issue comes to light. The combustion chamber pressure experiences a decline while the thruster is actively engaged in thrust production. As highlighted in the earlier discussion and evident in Figure 5, concerns about a system leakage have been identified. In alignment with anticipated complete combustion characteristics, the chamber pressure should demonstrate an increase during the burn. This could also indicate that the system was operating in an overexpanded condition. Regrettably, the findings from the first run indicate instances of both unstable and incomplete combustion.

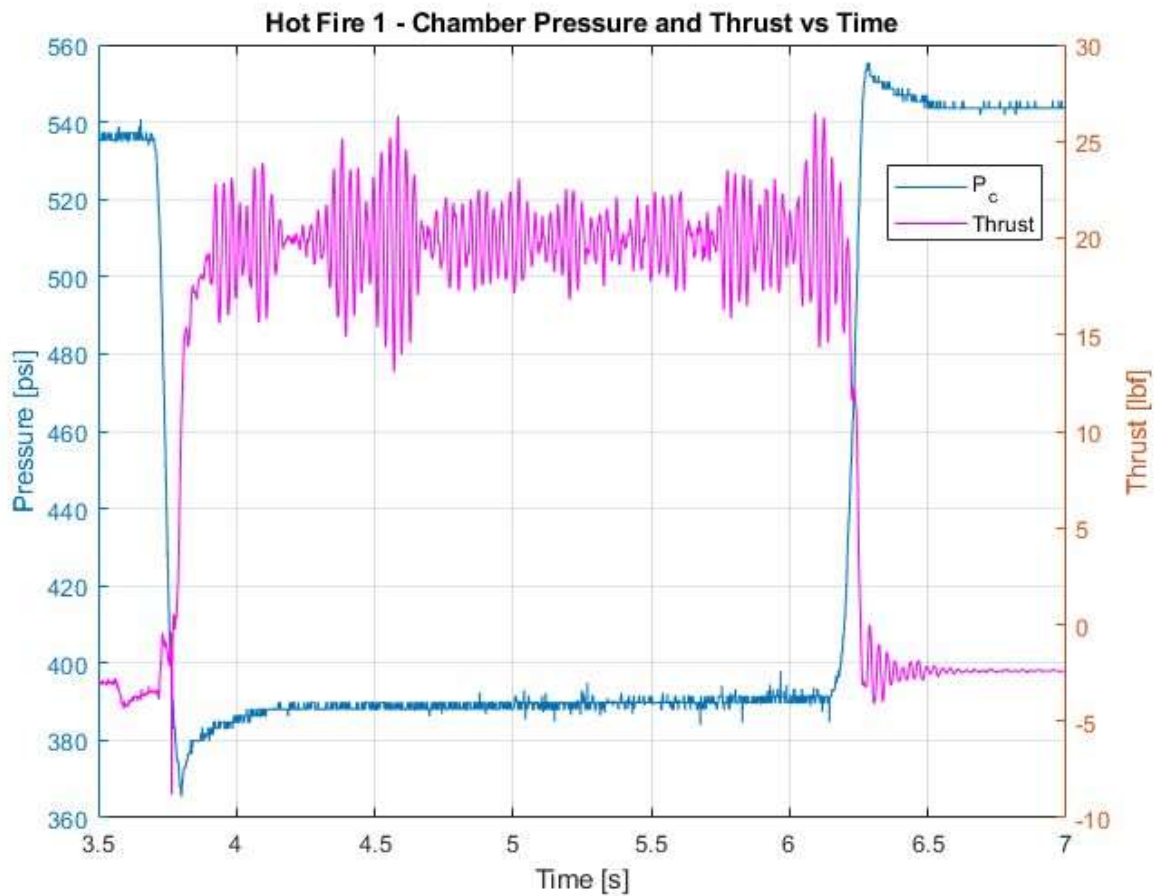


Fig. 6 First Run Chamber Pressure and Thrust Data.

In the second run, as depicted in both Figures 7 and 8, a consistent increase is observed across all pressure values between approximately 4.7 and 5.5 seconds. Throughout this time frame, the recorded pressures at the inlet (high pressure) and the throat (low pressure) consistently adhere to the principles of Bernoulli. Even as the velocity increases during the run, the pressures at the throat remain lower than those measured at the inlet, affirming the expected relationship dictated by Bernoulli's principle. Notably, the pressure values for ethanol and oxygen exhibit a correlated behavior during this period, contributing to the overall rise in combustion pressure. This positive correlation is further reflected in Figure 8, where a steady increase in thrust is evident within the same time frame.

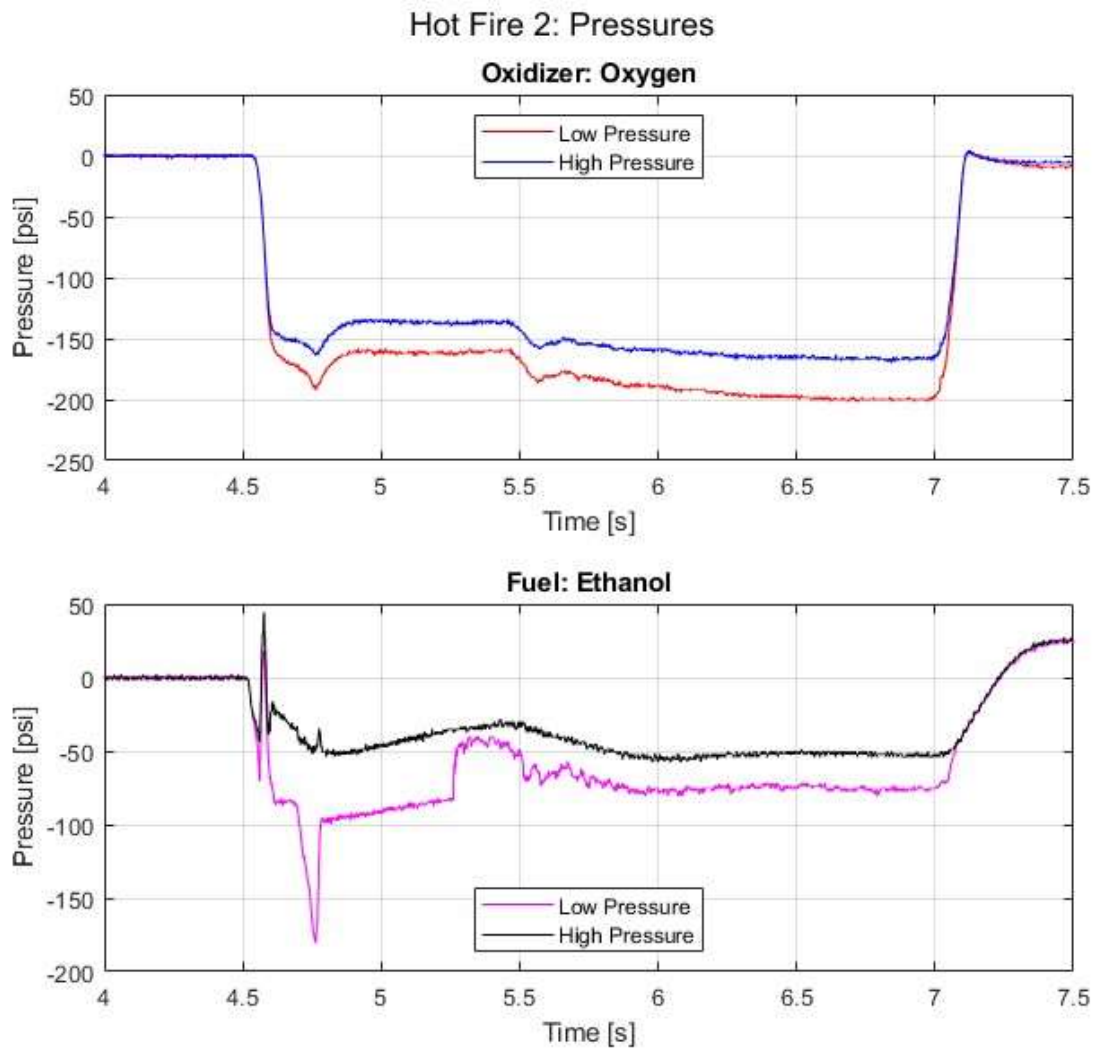


Fig. 7 Second Run Pressure Data of Propellant

However, subsequent to the 5.5-second mark, a noticeable decline is observed in both pressures and thrust, figure 8. This decline may signify changes in combustion dynamics or other factors influencing the propulsion system's performance. Further investigation is warranted to elucidate the specific causes behind the observed decrease in pressures and thrust beyond the 5.5-second time frame in order to optimize and refine the system for enhanced efficiency. In addition, it should be noted that the pressure data for hot fire 2 was tared incorrectly. The combustion chamber is not experiencing negative pressures in actuality. Furthermore, when adjusted similarly to the other pressure graphs, they are on similar magnitudes.

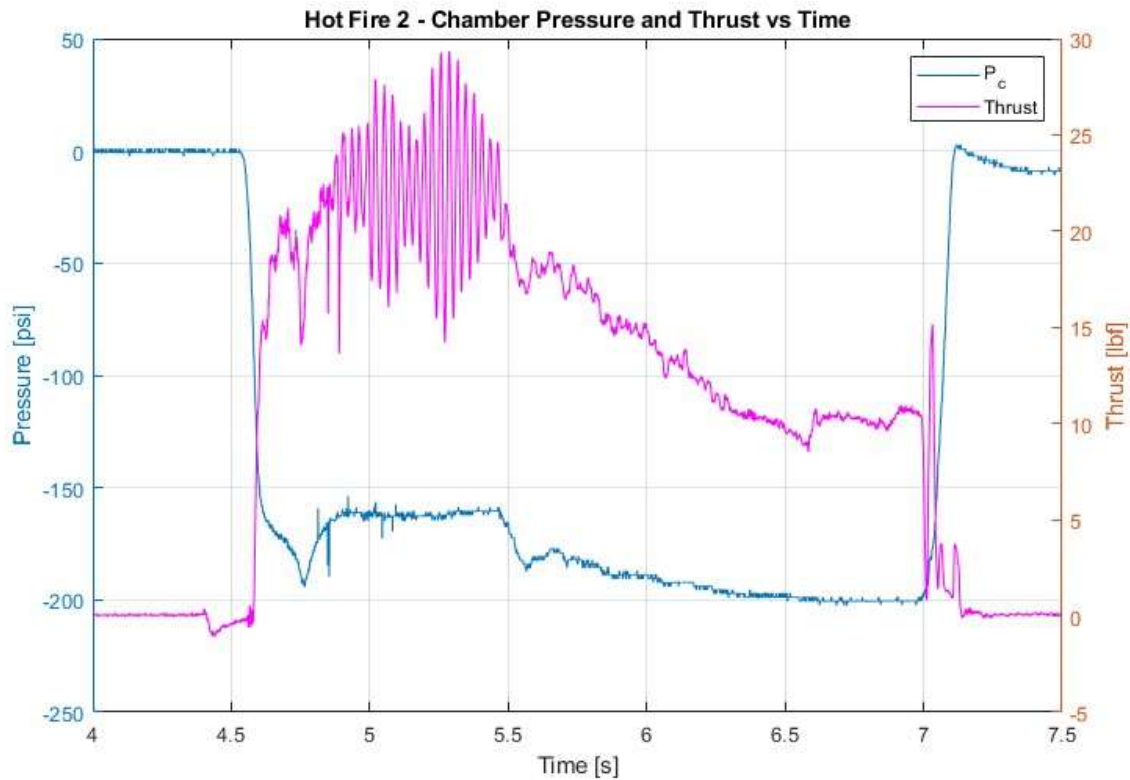


Fig. 8 Second Run Chamber Pressure and Thrust Data.

In the third run of the bi-propellant thruster, the oxidizer pressure exhibited a consistent pattern akin to the preceding two runs. However, a noteworthy deviation is observed in the ethanol pressures, as illustrated in Figure 9. At the onset of the burn, there is a drop in ethanol pressures. Intriguingly, the pressures swiftly revert to values around the initial readings between 4.5 and 5.7 seconds, suggesting a potential blockage in the ethanol inlet. Beyond the 5.7-second mark, a minor uptick in ethanol pressure is observed at the throat (low pressure), deviating from the previously established fluid dynamics principles. In Figure 9, noteworthy proximity between the low and high pressure values is apparent, suggesting a potential issue with combustion. According to expectations, the increase in velocity should result in a more substantial decrease in low-pressure values compared to high-pressure values.

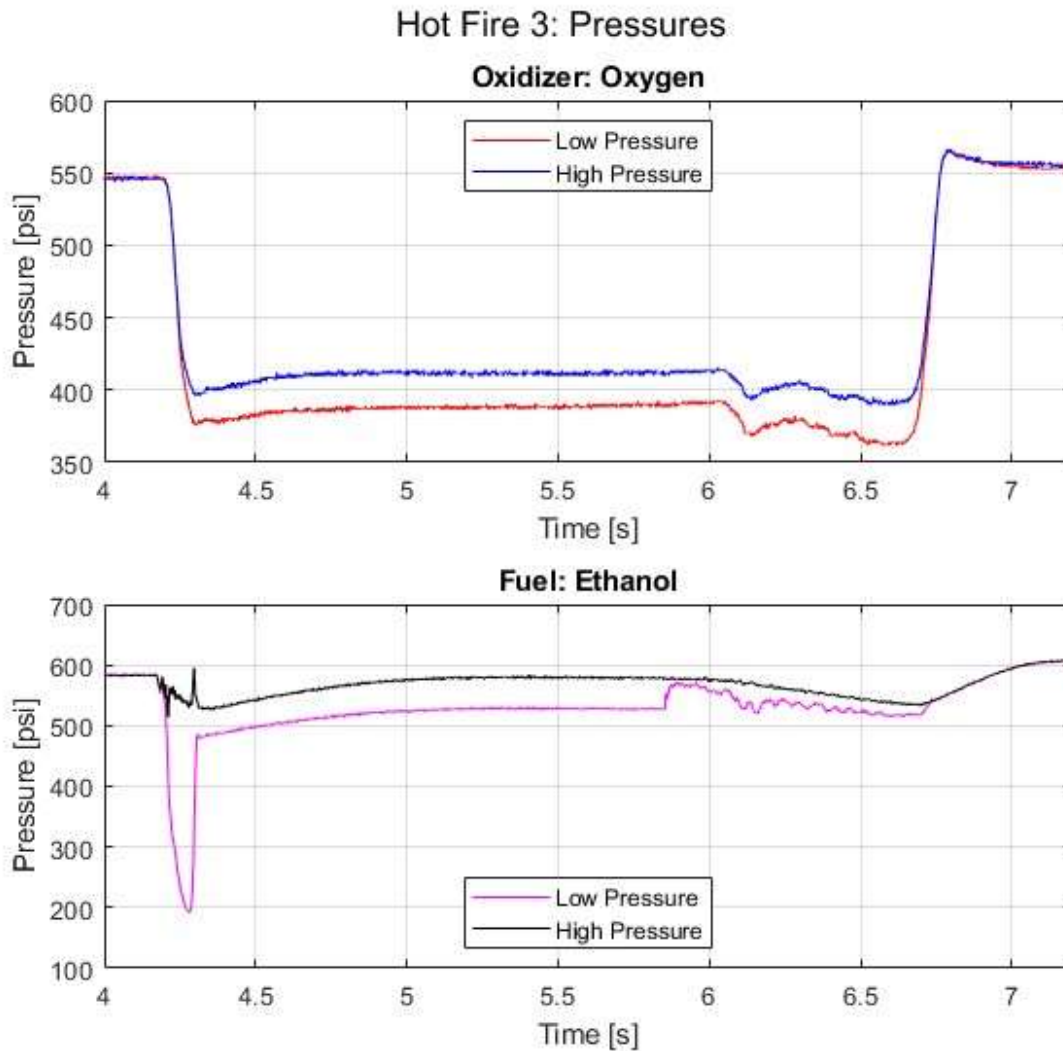


Fig. 9 Third Run Pressure Data of Propellant

In Figure 10, it is shown that the thrust attains higher magnitudes between 5.5 and 6.1 seconds, indicating partial combustion. However, this phase is short-lived, as the drop in oxidizer pressure and the close proximity of the ethanol pressures contributes to a subsequent decline in combustion pressure and thrust, ultimately resulting in the manifestation of unstable combustion.

For the achievement of optimal combustion efficiency in a bipropellant thruster, the graphical representation of the combustion process should illustrate a concurrent elevation in the plots depicting fuel, oxidizer, and chamber pressures throughout the burn duration. This observed increase in pressures substantiates a consistent and robust supply of propellants—fuel and oxidizer. The synchronized ascent in chamber pressure is indicative of a well-coordinated combustion process. This rise in pressure profiles collectively signifies an environment conducive to thorough mixing of fuel and oxidizer, thereby fostering efficient combustion kinetics and subsequently enhancing thrust output.

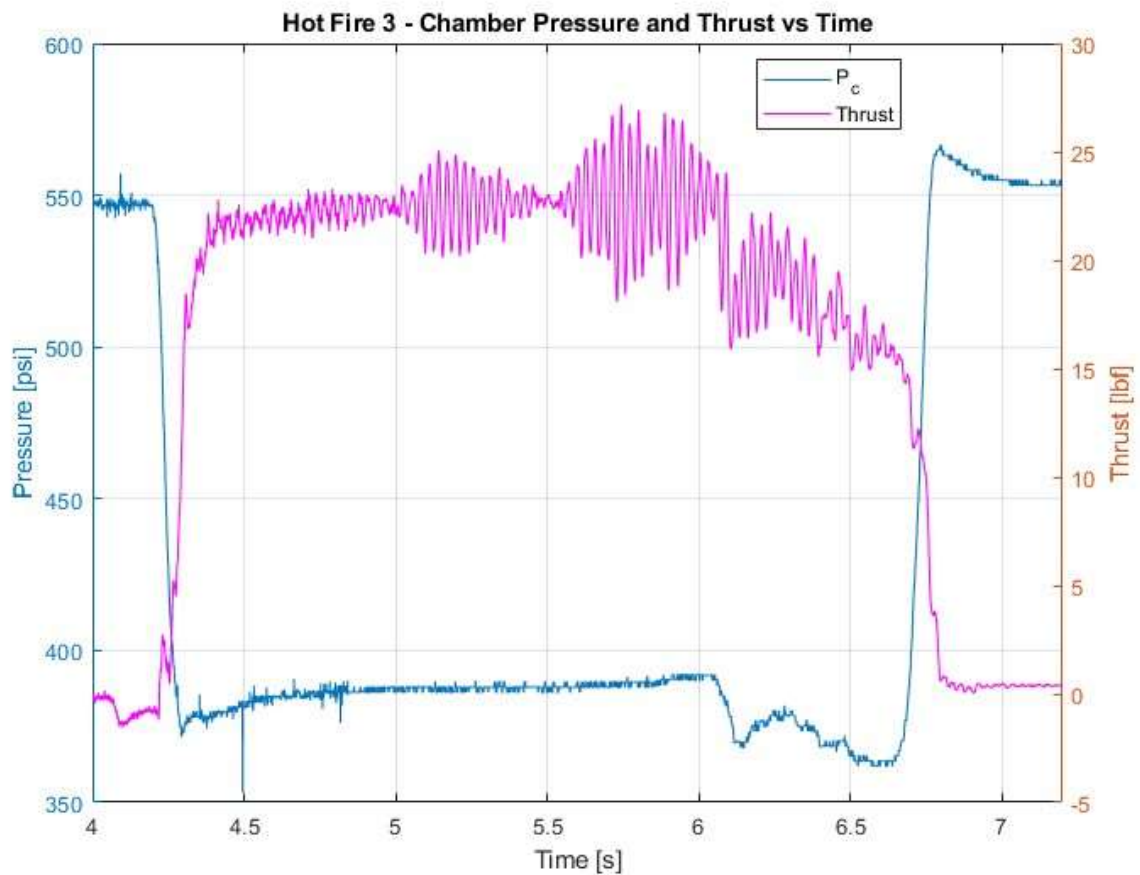


Fig. 10 Third Run Chamber Pressure and Thrust Data.

B. Performance Characteristics

The data presented in Table 1 outlines the characteristics and efficiency of the bipropellant thruster examined during the lab session. It is crucial to note that due to certain malfunctions observed in the thruster during the experiment, the data may not accurately represent the performance of a bipropellant thruster operating under conditions of complete combustion and efficiency.

In the Bipropellant Lab Manual, the mass flow rates for both oxidizer and fuel were initially specified based on the original design of the system—0.00767 slug/s for the oxidizer and 0.00366 slug/s for the fuel [1]. A comparison with the experimental data in Table 1 reveals significant deviations from these design values.

The discussion on pressure and thrust data, as detailed in the report, indicates a consistent drop in pressure during the runs. This phenomenon may be associated with the observed increase in mass flow rate. According to fluid dynamics principles, a drop in pressure, with other factors held constant, can result in fluctuations in fluid density, as per the ideal gas law. The elevated mass flow rates of both oxidizer and fuel might serve as a compensatory mechanism for incomplete combustion, representing an effort to supply more propellant to the combustion chamber and enhance efficiency.

The specific impulse values obtained from the runs are notably low, registering at less than ten seconds. Given that specific impulse reflects the efficiency in converting propellant into thrust, it is evident that this particular system demonstrated suboptimal efficiency. It is important to acknowledge that these results align with expectations, considering the multiple complications encountered during the lab session, particularly with regard to firing the system.

Table 1 Experimental Results

Runs	Burn Time	Avg. Thrust	Avg. Fuel Mass Flow Rate	Avg. O2 Mass Flow Rate	Mixture Ratio (O/F)	Isp
Hot Fire 1	2.9452 s	15.94 lbf	0.050159 slug/s	0.022396 slug/s	0.4465	6.8228 s
Hot Fire 2	2.7085 s	14.3885 lbf	0.037102 slug/s	0.026575 slug/s	0.71627	7.0175 s
Hot Fire 3	2.8214 s	18.3481 lbf	0.046361 slug/s	0.023877 slug/s	0.51501	8.1126 s

C. Balanced Chemical Equation

The thrust from the bi-propellant thruster is a results from the chemical reaction between gaseous oxygen (O_2) and liquid Ethanol (C_2H_5OH). This reaction yields carbon dioxide (CO_2) and water (H_2O) as products. Assuming complete combustion, the balanced chemical equation for this reaction is as follows:



Based on the chemical equation above, the stoichiometric mixture ratio should ideally be 2.09. Unfortunately, the experimental results indicate that the thruster didn't achieve complete combustion in any of the runs. Referring to Table 1, the average mixture ratio across the three runs is 0.55926, confirming the absence of complete combustion. As a result, the bi-propellant thruster did not achieve its maximum potential for thrust and efficiency.

D. Plume Observations

From the fig 11 of the exhaust plume, at some points in time we are clearly able to see visible shock diamonds. This is a strong visual cue that the exhaust from the engine is being over-expanded or under-expanded, as the shocks in the exhaust are a remedial natural measure to equalize the pressure of the exhaust and the back pressure (in this case, ambient pressure at near-sea level conditions). To determine whether it is over or under-expanded, we can look to the behavior of the plume at the nozzle exit. At the exit, we can see no "pinching" or compression of the plume; rather, we see that the plume expands without any compression after the exit, meaning that the exhaust pressure of the plume is greater than the ambient pressure surrounding the nozzle exit. This follows the trend of under-expanded exhaust plumes, as the plume will at first expand to a greater diameter than the nozzle exit, and later in the flow it will compress again. This can also be seen towards the tail-end of the exhaust plume in the image - after the shock diamonds have disappeared, the plume appears to become slightly more slim.



Fig. 11 Exhaust Plume with Shock Diamonds

E. Errors & Experiment Improvements

To increase the performance of the rocket nozzle, one important step would be to re-size the nozzle of the rocket to better suit it for atmospheric conditions at near-sea level elevation. Currently, the nozzle under-expands the exhaust, which means that a portion of the exhaust will be spread radially rather than axially out of the nozzle. This distribution of exhaust gases decreases our potential thrust, since our maximum thrust depends on ejecting as much mass in the direction opposite to the rocket as possible.

Several mechanical errors contributed to the reduced effectiveness of the rocket engine as well. The pressure-regulated cooling system was not working during the run, so an improvised cooling system using atmospheric pressure was devised. This cooling system was only marginally effective, as sometimes air bubbles would halt the flow of water through the piping, leaving the nozzle uncooled. This issue was resolved later, but still resulted in a weak cooling system that rapidly depleted. Additionally, the rocket engine itself was experiencing mechanical failures. Prior to the experiment, the rocket experienced external combustion, which likely signified some type of fuel leak. According to lab instructors, the o-rings that sealed the system were replaced, as well as the spark plug ignitor (which had been destroyed in the previous failure). Another issue was noted, although the cause is not certain: The rocket engine was experiencing trouble reaching full combustion in the combustion chamber. The exhaust plume exhibited signs of engine sputtering and discoloration, which might possibly indicate improper mixing ratios between the fuel and oxidizer.

F. Recommendations

After addressing potential performance concerns outlined in the analysis and error sections, it's important to take into account the recorded thrust. Based on this information, here are some potential recommendations for the specific application of this thruster. The average thrust for three runs is approximately 16.2 lbf. While bipropellants, as mentioned in the introduction, are extensively used in rocket propulsion, the scale of thrust required to overcome a rocket's weight is astronomically larger than 16.2 lbf. Consequently, this specific bipropellant thruster may find potential applications in missions that necessitate smaller thrust. One prominent example is in the context of attitude control for satellites operating in the vacuum of space. By combining several of these thrusters based on mission requirements, a larger net thrust force can be generated, enabling in-space maneuvering of larger space vehicles.

V. Conclusion

In summary, this lab experiment provided a comprehensive exploration into the intricacies of bipropellant thruster utilization, uncovering a range of challenges that impact its characteristics. Despite meticulous execution, issues in data acquisition, from calibration errors to combustion occurrences outside the chamber, alongside unexpected pressure fluctuations, were revealed. Nevertheless, the experiment yielded unexpected insights into characteristics and trends not foreseen for data derived from a bipropellant thruster assuming complete combustion. The data indicated a failure to achieve complete combustion, evident in O/F ratios of 0.4465, 0.71627, and 0.51501, significantly deviating from the stoichiometric mixture ratio of 2.09, assuming complete combustion. These discrepancies underscore the non-ideal combustion conditions observed in the experiment. Plume observations, particularly the presence of visible shock diamonds, indicate over- or under-expansion, aligning with incomplete combustion. The lack of compression in the plume after the exit further suggests under-expanded exhaust plumes, contributing to the experiment's complexities. While the experiment faced challenges, it has provided invaluable insights that extend beyond the anticipated outcomes. The identified issues serve as a foundation for future optimizations and advancements in bipropellant thruster technology.

References

- 1) Saucier, William . “Bipropellant Thruster.” CalPoly Canvas, California Polytechnic State University, San Luis Obispo
- 2) Saucier, William . “Biprop Firing Procedure.” CalPoly Canvas, California Polytechnic State University.
- 3) SpaceX. “Falcon 9.” SpaceX, 2021, www.spacex.com/vehicles/falcon-9/.
- 4) “Engines.” Blue Origin, www.blueorigin.com/engines. Accessed 4 Nov. 2023.

Appendix

MATLAB Code: Attached is the MATLAB code used for data analysis and processing in this experiment. The code provides a detailed implementation of the procedures involved in refining and interpreting the acquired data

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```
% Aero 402
% BiProp Lab
```

```
clc
clear all
```

```
% Group A.2
```

Upload data

```
% Hot Fire 1
HF_1 = xlsread('HotFire_1.xlsx');

% Access specific elements
X_Value = HF_1(:,1); % seconds
Engine_DuctCellLoad = HF_1(:,2);
Eth_Ven_HP = HF_1(:,4); % psi
Eth_Ven_LP = HF_1(:,5); % psi
Oxy_Ven_HP = HF_1(:,6); % psi
Oxy_Ven_LP = HF_1(:,7); % psi
Pc = HF_1(:,8); % psi
```

Pressure & Thrust Plots

```
% Oxygen preesure plot
figure()
subplot(2,1,1)
plot(X_Value,Oxy_Ven_LP,'r',X_Value,Oxy_Ven_HP,'b')
legend("Low Pressure","High Pressure","Location",'North')
title('Oxidizer: Oxygen')
ylabel("Pressure [psi]")
xlabel('Time [s]')
xlim([3.5 7])
grid on
axis 'auto y'

% Ethanol Pressure Plot
```

```

subplot(2,1,2)
plot(X_Value,Eth_Ven_LP,'m',X_Value,Eth_Ven_HP,'k')
legend("Low Pressure","High Pressure",'Location','South')
title('Fuel: Ethanol')
ylabel("Pressure [psi]")
xlabel('Time [s]')
xlim([3.5 7])
grid on
axis 'auto y'
sgtitle("Hot Fire 1: Pressures ")

```

```

% Plot Pressure on the left y-axis
figure()
yyaxis left
plot(X_Value, Pc)
ylabel('Pressure [psi]')
grid on
% Adding second y-axis for Thrust
yyaxis right
plot(X_Value, Engine_DuctCellLoad, 'm')
ylabel('Thrust [lbf]')
legend('P_c','Thrust')
grid on
title('Hot Fire 1 - Chamber Pressure and Thrust vs Time')
xlabel('Time [s]')
xlim([3.5 7])

```

Refine Burn Times

```

% Observed values from Thrust pressure vs Time plots
startBurnTime = 3.5918; % seconds
endBurnTime = 6.537; % seconds

% Find indices corresponding to the specified time range
startBurnIndex = find(X_Value >= startBurnTime, 1);
endBurnIndex = find(X_Value <= endBurnTime, 1, 'last');

% Extract the burn time data
burnTimeData_X = X_Value(startBurnIndex:endBurnIndex);
burnTimeData_Thrust = Engine_DuctCellLoad(startBurnIndex:endBurnIndex);
burnTimeData_Eth_LP = Eth_Ven_LP(startBurnIndex:endBurnIndex);
burnTimeData_Eth_HP = Eth_Ven_HP(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_LP = Oxy_Ven_LP(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_HP = Oxy_Ven_HP(startBurnIndex:endBurnIndex);

% Calculate the burn time in seconds
burnTimeInSeconds = endBurnTime - startBurnTime;

% Display the burn time
disp(['Burn Time: ' num2str(burnTimeInSeconds) ' seconds']);

```

Ave Thrust

```
ave_T1 = mean(burnTimeData_Thrust);  
  
disp(['Average Thrust during Burn 1 is: ' num2str(ave_T1) ' lbf']);
```

Ave volumetric fuel Flow rate

```
% Assume incompressible flow in venturi  
% Area of cross section inches  
A_t_eth = (pi*(0.070/24)^2); % from in to ft  
A_in_eth = (pi*(0.248/24)^2);  
% Density of fuel  
rho_eth = 49.25/32.2; % slug/ft^3  
  
% Pressure diff [psi]  
Eth_LP = mean(burnTimeData_Eth_LP); % Throat  
Eth_HP = mean(burnTimeData_Eth_HP); % Inlet  
dp_Eth = ((Eth_HP-Eth_LP)*144); % lbf/ft^2  
  
ave_V = sqrt((2*dp_Eth)/rho_eth); %ft/s  
  
vdot_f = A_in_eth*ave_V; % ft^3/s  
mdot_f = (rho_eth*vdot_f); % slug/s  
disp("Ethanol: "+mdot_f+" slug/s")
```

Ave Oxygen mass flow rate

```
% Area of cross section [ft]^2  
A_t_oxy = pi*(0.125/24)^2;  
A_in_oxy = pi*(0.248/24)^2;  
% density of oxy  
rho_oxy = 27.23/32.2; % slug/ft^3  
% Assume incompressible flow in venturi  
  
% Pressure diff  
Oxy_LP = mean(burnTimeData_Oxy_LP); % Throat  
Oxy_HP = mean(burnTimeData_Oxy_HP); % Inlet  
dp_Oxy = (Oxy_HP-Oxy_LP)*144; % lbf/ft^2  
  
ave_V_Oxy = sqrt((2*dp_Oxy)/rho_oxy); %ft/s;  
  
vdot_o = A_in_oxy*ave_V_Oxy; % ft^3/s  
mdot_o = (rho_oxy*vdot_o); % slug/s  
disp("Oxygen: "+mdot_o+" slug/s")
```

O/F

```
OF = (mdot_o/burnTimeData_X)/(mdot_f/burnTimeData_X);  
disp("O/F: "+OF)
```

I_sp

```
% Total mass flow rate
mdot = mdot_o + mdot_f;

% turn lbf into slug
Isp = (ave_T1/32.2)/(mdot);

disp("Specific Impulse: "+Isp+"seconds.")
```

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Aero 402

```
% Hot Fire 2 Data Calcs Plots
clc
clear all

% Group A.2
```

Upload data

```
% Hot Fire 2
HF_2 = xlsread('HotFire_2.xlsx');

% Access specific elements
X_Value2 = HF_2(:,1); % seconds
Engine_DuctCellLoad2 = HF_2(:,2);
Eth_Ven_HP2 = HF_2(:,4);
Eth_Ven_LP2 = HF_2(:,5);
Oxy_Ven_HP2 = HF_2(:,6);
Oxy_Ven_LP2 = HF_2(:,7);
Pc2 = HF_2(:,8);
```

Pressure vs Thrust Plots

```
% Oxygen preesure plot
figure()
subplot(2,1,1)
plot(X_Value2,Oxy_Ven_LP2,'r',X_Value2,Oxy_Ven_HP2,'b')
legend("Low Pressure","High Pressure","Location","North")
title('Oxidizer: Oxygen')
ylabel("Pressure [psi]")
xlabel('Time [s]')
xlim([4 7.5])
grid on

% Ethanol Pressure Plot
subplot(2,1,2)
```

```

plot(X_Value2,Eth_Ven_LP2,'m',X_Value2,Eth_Ven_HP2,'k')
legend("Low Pressure","High Pressure",'Location','South')
title('Fuel: Ethanol')
ylabel("Pressure [psi]")
xlabel('Time [s]')
xlim([4 7.5])
grid on
sgtitle("Hot Fire 2: Pressures ")

```

```

% Plot Pressure on the left y-axis
figure()
yyaxis left
plot(X_Value2, Pc2)
ylabel('Pressure [psi]')
grid on
% Adding second y-axis for Thrust
yyaxis right
plot(X_Value2, Engine_DuctCellLoad2, 'm')
ylabel('Thrust [lbf]')
legend('P_c','Thrust')
grid on
title('Hot Fire 2 - Chamber Pressure and Thrust vs Time')
xlabel('Time [s]')
xlim([4 7.5])

```

Refine Burn Times

```

% Observed values from Thrust pressure vs Time plots
startBurnTime = 4.4299; % seconds
endBurnTime = 7.1384; % seconds

% Find indices corresponding to the specified time range
startBurnIndex = find(X_Value2 >= startBurnTime, 1);
endBurnIndex = find(X_Value2 <= endBurnTime, 1, 'last');

% Extract the burn time data
burnTimeData_X = X_Value2(startBurnIndex:endBurnIndex);
burnTimeData_Thrust = Engine_DuctCellLoad2(startBurnIndex:endBurnIndex);
burnTimeData_Eth_LP = Eth_Ven_LP2(startBurnIndex:endBurnIndex);
burnTimeData_Eth_HP = Eth_Ven_HP2(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_LP = Oxy_Ven_LP2(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_HP = Oxy_Ven_HP2(startBurnIndex:endBurnIndex);

% Calculate the burn time in seconds
burnTimeInSeconds = endBurnTime - startBurnTime;

% Display the burn time
disp(['Burn Time: ' num2str(burnTimeInSeconds) ' seconds']);

```

Ave Thrust

```
ave_T2 = mean(burnTimeData_Thrust);  
  
disp(['Average Thrust during Burn 2 is: ' num2str(ave_T2) ' lbf']);
```

Ave fuel Flow rate

```
% Assume incompressible flow in venturi  
% Area of cross section inches  
A_t_eth = (pi*(0.070/24)^2); % from in to ft  
A_in_eth = (pi*(0.248/24)^2);  
% Density of fuel  
rho_eth = 49.25/32.2; % slug/ft^3  
  
% Pressure diff [psi]  
Eth_LP = mean(burnTimeData_Eth_LP); % Throat  
Eth_HP = mean(burnTimeData_Eth_HP); % Inlet  
dp_Eth = ((Eth_HP-Eth_LP)*144); % lbf/ft^2  
  
ave_V = sqrt((2*dp_Eth)/rho_eth); %ft/s  
  
vdot_f = A_in_eth*ave_V; % ft^3/s  
mdot_f = (rho_eth*vdot_f); % slug/s  
disp("Ethanol: "+mdot_f+" slug/s")
```

Ave Oxygen mass flow rate

```
% Area of cross section [ft]^2  
A_t_oxy = pi*(0.125/24)^2;  
A_in_oxy = pi*(0.248/24)^2;  
% density of oxy  
rho_oxy = 27.23/32.2; % slug/ft^3  
% Assume incompressible flow in venturi  
  
% Pressure diff  
Oxy_LP = mean(burnTimeData_Oxy_LP); % Throat  
Oxy_HP = mean(burnTimeData_Oxy_HP); % Inlet  
dp_Oxy = (Oxy_HP-Oxy_LP)*144; % lbf/ft^2  
  
ave_V_Oxy = sqrt((2*dp_Oxy)/rho_oxy); %ft/s;  
  
vdot_o = A_in_oxy*ave_V_Oxy; % ft^3/s  
mdot_o = (rho_oxy*vdot_o); % slug/s  
disp("Oxygen: "+mdot_o+" slug/s")
```

O/F

```
OF = (mdot_o/burnTimeData_X)/(mdot_f/burnTimeData_X);  
disp("O/F: "+OF)
```

I_sp

```
% Total mass flow rate
mdot = mdot_o + mdot_f;

% turn lbf into slug
Isp = (ave_T2/32.2)/(mdot);

disp("Specific Impulse: "+Isp+"seconds.")
```

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Aero 402

```
% Biprop Lab
% Hot Fire 3
clc
clear all
```

```
% Group A.2
```

Upload data

```
% Hot Fire 3
HF_3 = xlsread('HotFire_3.xlsx');

% Access specific elements
X_Value3 = HF_3(:,1); % seconds
Engine_DuctCellLoad3 = HF_3(:,2);
Eth_Ven_HP3 = HF_3(:,4);
Eth_Ven_LP3 = HF_3(:,5);
Oxy_Ven_HP3 = HF_3(:,6);
Oxy_Ven_LP3 = HF_3(:,7);
Pc3 = HF_3(:,8);
```

Pressure vs Thrust Plots

```
% Oxygen pressure plot
figure()
subplot(2,1,1)
plot(X_Value3,Oxy_Ven_LP3,'r',X_Value3,Oxy_Ven_HP3,'b')
legend('Low Pressure','High Pressure','Location','North')
title('Oxidizer: Oxygen')
ylabel('Pressure [psi]')
xlabel('Time [s]')
xlim([4 7.2])
grid on
axis 'auto y'
```

```

% Ethanol Pressure Plot
subplot(2,1,2)
plot(X_Value3,Eth_Ven_LP3,'m',X_Value3,Eth_Ven_HP3,'k')
legend("Low Pressure","High Pressure",'Location','South')
title('Fuel: Ethanol')
ylabel("Pressure [psi]")
xlabel('Time [s]')
xlim([4 7.2])
grid on
axis 'auto y'
sgtitle("Hot Fire 3: Pressures ")

```

```

% Plot Pressure on the left y-axis
figure()
yyaxis left
plot(X_Value3, Pc3)
ylabel('Pressure [psi]')
grid on
% Adding second y-axis for Thrust
yyaxis right
plot(X_Value3, Engine_DuctCellLoad3, 'm')
ylabel('Thrust [lbf]')
legend('P_c','Thrust')
grid on
title('Hot Fire 3 - Chamber Pressure and Thrust vs Time')
xlabel('Time [s]')
xlim([4 7.2])

```

Refine Burn Times

```

% Observed values from Thrust pressure vs Time plots
startBurnTime = 4.0891; % seconds
endBurnTime = 6.9105; % seconds

% Find indices corresponding to the specified time range
startBurnIndex = find(X_Value3 >= startBurnTime, 1);
endBurnIndex = find(X_Value3 <= endBurnTime, 1, 'last');

% Extract the burn time data
burnTimeData_X = X_Value3(startBurnIndex:endBurnIndex);
burnTimeData_Thrust = Engine_DuctCellLoad3(startBurnIndex:endBurnIndex);
burnTimeData_Eth_LP = Eth_Ven_LP3(startBurnIndex:endBurnIndex);
burnTimeData_Eth_HP = Eth_Ven_HP3(startBurnIndex:endBurnIndex);
burnTimeData_Eth_HP = Eth_Ven_HP3(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_LP = Oxy_Ven_LP3(startBurnIndex:endBurnIndex);
burnTimeData_Oxy_HP = Oxy_Ven_HP3(startBurnIndex:endBurnIndex);

% Calculate the burn time in seconds
burnTimeInSeconds = endBurnTime - startBurnTime;

% Display the burn time
disp(['Burn Time: ' num2str(burnTimeInSeconds) ' seconds']);

```

Ave Thrust

```
ave_T3 = mean(burnTimeData_Thrust);  
  
disp(['Average Thrust during Burn 3 is: ' num2str(ave_T3) ' lbf']);
```

Ave volumetric fuel Flow rate

```
% Assume incompressible flow in venturi  
% Area of cross section inches  
A_t_eth = (pi*(0.070/24)^2); % from in to ft  
A_in_eth = (pi*(0.248/24)^2);  
% Density of fuel  
rho_eth = 49.25/32.2; % slug/ft^3  
  
% Pressure diff [psi]  
Eth_LP = mean(burnTimeData_Eth_LP); % Throat  
Eth_HP = mean(burnTimeData_Eth_HP); % Inlet  
dp_Eth = ((Eth_HP-Eth_LP)*144); % lbf/ft^2  
  
ave_V = sqrt((2*dp_Eth)/rho_eth); %ft/s  
  
vdot_f = A_in_eth*ave_V; % ft^3/s  
mdot_f = (rho_eth*vdot_f); % slug/s  
disp("Ethanol: "+mdot_f+" slug/s")
```

Ave Oxygen mass flow rate

```
% Area of cross section [ft]^2  
A_t_oxy = pi*(0.125/24)^2;  
A_in_oxy = pi*(0.248/24)^2;  
% density of oxy  
rho_oxy = 27.23/32.2; % slug/ft^3  
% Assume incompressible flow in venturi  
  
% Pressure diff  
Oxy_LP = mean(burnTimeData_Oxy_LP); % Throat  
Oxy_HP = mean(burnTimeData_Oxy_HP); % Inlet  
dp_Oxy = (Oxy_HP-Oxy_LP)*144; % lbf/ft^2  
  
ave_V_Oxy = sqrt((2*dp_Oxy)/rho_oxy); %ft/s;  
  
vdot_o = A_in_oxy*ave_V_Oxy; % ft^3/s  
mdot_o = (rho_oxy*vdot_o); % slug/s  
disp("Oxygen: "+mdot_o+" slug/s")
```

O/F

```
OF = (mdot_o/burnTimeData_X)/(mdot_f/burnTimeData_X);  
disp("O/F: "+OF)
```

I_sp

```
% Total mass flow rate
mdot = mdot_o + mdot_f;

% turn lbf into slug
Isp = (ave_T3/32.2)/(mdot);

disp("Specific Impulse: "+Isp+"seconds.")
```

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